

Throughout my academic career, I have contested with the idea of privacy and security through technical and human-centered, justice-oriented research, converging on a central question: how do technologies enable abusers to exploit power & control and perpetuate interpersonal and structural violence, and how can private and secure design help communities reclaim safety, healing, and liberation?

I began my technical training with a Bachelor of Technology in Computer Science & Engineering from the Indraprastha Institute of Information Technology, Delhi (2013 to 2017), where my coursework and undergraduate thesis introduced me to systems security research. After graduating, I spent several years in the industry to utilize my security skills to real-world applications. As a DevSecOps Engineer at Grab, across India and Singapore, I worked on technical projects to protect private sensitive data at scale for Grab's payment infrastructure. I implemented application security policies across company-wide Kubernetes clusters and designed secret-sharing protocols for third-party data exchanges with policymakers. This industry experience led me to pursue a Research Fellowship at Microsoft Research India (2020–2021), where I worked across the Technology for Empowerment group and the Confidential Computing Group. I designed a privacy-preserving COVID-19 health certificate issuance system built on Microsoft's Confidential Computing Framework, conducted security evaluations of Trusted Execution Environments for GPUs, and developed a toolchain to evaluate one of the first formally verified secure boot implementations on commodity hardware. These industry roles sharpened my interest in technical privacy engineering against canonical adversarial threat models.

At the University of Wisconsin–Madison (2021-present), I went through my own experiences of violence in my personal and professional life from my ex-advisor through a long journey to find healing and resilience. As a survivor and an immigrant, I resolved to reciprocate my knowledge and experiences of support back to the gender-based violence survivors. Embracing an *advocacy-researcher* practice, I got involved as a consultant at the Madison Tech Clinic (MTC). Now, I direct the operations and outreach at MTC, a trauma-informed intervention that has supported 180+ survivors with their digital safety needs. I have personally supported 40+ survivors and lead the outreach efforts to build partnerships with 24 GBV advocacy organizations, spreading MTC's services across 32 counties in the state of Wisconsin, US. As a first-generation Computer Science PhD student, I had the opportunity to take interdisciplinary courses for my minor spanning Psychology, Gender & Women's Studies, Law, and Education Policy; courses which are rarely accessible in a computer science program in India. Such a interdisciplinary education and research collaboration allowed me to view privacy more broadly that included violations where the adversary and the victim are related through their physical proximity, social identities, & power dynamics. Lately, my research also engages how these oppressive logics and desire for privacy and safety intersect with AI. My work ranged from studying AI-mediated video conferencing experiences for blind and low-vision presenters in high-stakes meetings, to examining how AI data infrastructure intersects with settler-colonial harm, a topic that I recently presented at the Heidelberg Laureate Forum's NextGen session in 2026.

Amid scientific acceleration, BigTech AI systems introduce new privacy crises, enabling surveillance and extraction of resources. I use sensitive social justice theories to inform the design of AI towards transparency, responsibility, and self-determination. I am committed to designing interventions informed by the realities of the people most affected by its failures by collaborating directly with impacted communities, advocates, and policymakers.